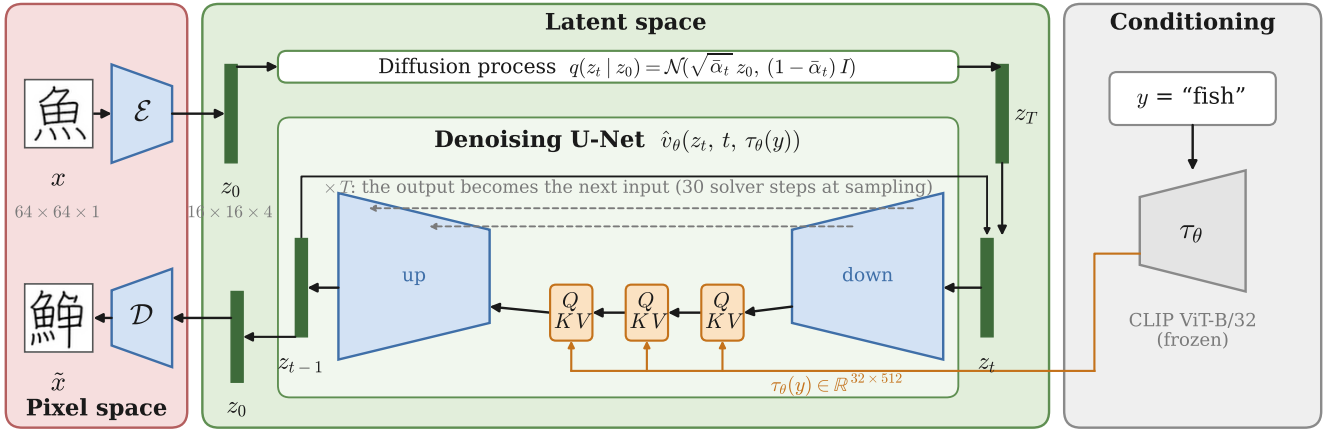

Latent space
Diffusion process $q(z_t \mid z_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\, z_0,\ (1-\bar{\alpha}_t)\, I)$
Denoising U-Net $\hat{v}_\theta(z_t,\ t,\ \tau_\theta(y))$
$\times T$: the output becomes the next input (30 solver steps at sampling)
z_T
z_0
$16 \times 16 \times 4$
up
down
Q
KV
z_{t-1}
z_t
z_0
$\tau_\theta(y) \in \mathbb{R}^{32 \times 512}$
$\mathcal{E}$
x
$64 \times 64 \times 1$
$\mathcal{D}$
$\tilde{x}$
Pixel space
Conditioning
y = “fish”
τ_θ
CLIP ViT-B/32
(frozen)
cross-attention
skip connection
trained
frozen
latent tensor